

# Sara Zhiliu He . Jarvis Consulting

I earned a Master's degree in Systems Design Engineering (Artificial Intelligence and Machine Learning Specialization) from the University of Waterloo in October 2024, where I built great foundation for software engineering and AI. Eager to expand my skill set, I completed a software engineering diploma program at BrainStation in February 2025, gaining hands-on experience in full-stack web development. I am an energetic software developer with a solid foundation in data structures, algorithms, programming languages (Java, JavaScript, Python), databases, and a passion for learning and applying new technologies to solve complex problems and build innovative solutions.

## Skills

**Proficient:** Java, Spring Boot, Linux/Bash, RDBMS/SQL, JavaScript/TypeScript, React, Python, Agile/Scrum, Git

**Competent:** Angular, Node.js/Express, MongoDB, CI/CD, Docker, Pandas/NumPy

**Familiar:** Pattern Recognition, TensorFlow, PyTorch, Scikit-Learn, Figma

## Jarvis Projects

Project source code: [https://github.com/jarviscanada/jarvis\\_data\\_eng\\_SarazhiliuHe](https://github.com/jarviscanada/jarvis_data_eng_SarazhiliuHe)

**Linux Cluster Monitoring Agent** [GitHub]: Built the Jarvis Linux Cluster Monitoring Agent to assist the Linux Cluster Administration (LCA) team in managing a multi-node Rocky Linux environment. Implemented a Bash Script to collect static system information, such as CPU architecture, memory, and disk specs, executed once on each node. Automated the collection of real-time resource usage data, such as CPU, memory, and disk usage, using a Bash script triggered by a Cron job every minute. Deployed a PostgreSQL database in a Docker container to store the collected data in a portable environment. Used Git and GitHub for source code management and version control.

**Core Java Apps** [GitHub]:

- Twitter App: Curabitur laoreet tristique leo, eget suscipit nisi. Sed in sodales ex. Maecenas vitae tincidunt dui, et eleifend quam.
- JDBC App: Curabitur laoreet tristique leo, eget suscipit nisi. Sed in sodales ex. Maecenas vitae tincidunt dui, et eleifend quam.
- Grep App: Curabitur laoreet tristique leo, eget suscipit nisi. Sed in sodales ex. Maecenas vitae tincidunt dui, et eleifend quam.

**Springboot App** [GitHub]: Not Started

**Python Data Analytics** [GitHub]: Not Started

**Hadoop** [GitHub]: Not Started

**Spark** [GitHub]: Not Started

**Cloud/DevOps** [GitHub]: Not Started

## Highlighted Projects

**MasterMind** [GitHub]: Developed and maintained a web application for educational content creators using Next.js and TypeScript, and created reusable UI components to enhance consistency and maintainability. Designed and optimized PostgreSQL databases using Prisma ORM for seamless and efficient data interactions. Implemented security best practices, including role-based access control, and integrated secure authentication with Clerk. Deployed the application on Vercel, ensuring fast load times and global availability.

**InStock:** Built a responsive warehouse inventory management website using HTML, CSS, React, and JavaScript, adhering to functional and visual design requirements. Developed and managed APIs with Node.js, Express.js, and Knex.js, optimizing database interactions with MySQL. Handled version control and team collaboration using Git and GitHub, utilized Jira for project management, and participated in code reviews to ensure high-quality development and on-time delivery.

## Professional Experiences

**Software Developer, Jarvis (APR 2025-present):** Developed a Linux Cluster Monitoring Agent using Bash, PostgreSQL, and Docker to track and store system and resource usage data across multiple nodes. Designed relational database schemas and implemented queries in PostgreSQL to support structured data processing and reporting. Built Java applications focused on object-oriented design, data structures, and algorithm implementation.

**Full Stack Software Developer, Skill4Skill (MAR 2025-present):** Collaborated with cross-functional teams to develop Skill4Skill, a platform where individuals can exchange expertise, using TypeScript, React, Node.js and MongoDB for data management. Worked with designers and project managers to improve user experiences, and used Jira for user story development and backlog management. Maintained documentation for codebases and workflows and adapted content based on product updates and user feedback.

**Research Assistant, University of Waterloo (MAY 2024-DEC 2024):** Researched and developed supply chain optimization models with Python, Pandas, Numpy and Gurobi. Documented findings in analytical reports, communicated issues and solutions effectively with the supervising professor, and refined the paper based on feedback.

## Education

**University of Waterloo (2023-2024),** Master of Engineering, Systems Design Engineering - GPA: 85/100

**South China University of Technology (2019-2023),** Bachelor of Engineering, Logistics Engineering - GPA: 92/100  
- First-class Academic Scholarship - Student Science and Technology Innovation Competition Scholarship - PG Logistics Scholarship

## Miscellaneous

- BrainStation Software Engineering Diploma (2024)
- Balsillie School of International Affairs Technology Governance Summer School (2024)
- First prize in the National Contest in Big Data Analysis and Mining
- Third Prize in the China Undergraduate Mathematical Contest in Modeling
- Chinese Classical Dance (Captain of Dance Team at South China University of Technology)
- Volunteer Experience in Environmental and Community Service